

- 1 (a) (i) product of force and distance moved
(by force) in the direction of the force M1
A1 [2]
- (ii) work (done) per unit time (*idea of ratio needed*) B1 [1]
- (b) *either* work/time or power = (force $\times$ distance)/time M1
to give power = force $\times$ velocity A1 [2]
- (c) (i) kinetic energy ($= \frac{1}{2}mv^2$) = $\frac{1}{2} \times 1900 \times 27^2$ C1
power = $692550 / 8.1 = 8.55 \times 10^4$ W A1 [2]
- (ii) *either* for equal increments of speed, increments of E_K are different M1
so longer time (to increase speed) at high speeds A1 [2]
- or* air resistance increases with speed (M1)
so driving force (and acceleration) reduced (A1)
- or* $P (= Fv) = mav$ (M1)
(P and m constant) so when v increases, a decreases (A1)

- 2 (a) uses a tangent (anywhere), not a single point C1